Delivered tables — preview

Not a deliverable. Layout here is arbitrary; the paper's floats are D's call. Generated by `python -m analysis.preview`*.*

table1_dataset

Statistic	Development	Test
Paragraphs	2000	2000
duplicated across splits	1	1
Source reports (PDFs)	49	49
shared across splits	49	49
Companies	50	50
spanning >1 report	0	n/a
Legal states observed	15 / 17	n/a
Rarest classes		
within_2_years	34	n/a
Misleading	2	n/a

Dataset and split statistics, regenerated from the versioned data by analysis/audit.py. Misleading occurs twice in the whole development set, in two different reports, and is absent from the Calibration partition of 18 of the 30 rotations. All 49 development reports also appear in the test split, and one paragraph text is duplicated across the two. No company contributes more than one report, so a document-disjoint split is also company-disjoint. The competition test split ships no labels, so its label-derived cells are marked n/a.

table2_main

ID	Calibration	Decoding	Weighted F1	PS	VT	ES	EQ	Tuple Acc.	Invalid %
M0	None	Independent	0.572±0.003	0.796	0.464	0.650	0.424	0.359	12.6
M1	None	Projection	0.571±0.003	0.796	0.467	0.651	0.418	0.394	0.0
M2	Global	Projection	0.574±0.004	0.796	0.493	0.650	0.418	0.428	0.0
M3	Conditional	Projection	0.574±0.005	0.796	0.501	0.651	0.412	0.431	0.0
M4	None	17-state	0.571±0.005	0.799	0.468	0.648	0.420	0.388	0.0
M5	Global	17-state	0.576±0.001	0.796	0.493	0.649	0.423	0.430	0.0
M6	Conditional	17-state	0.567±0.009	0.784	0.494	0.636	0.414	0.426	0.0

Controlled comparison of decision rules on identical base probabilities and identical test rows. Each cell is the mean over three seeds of a single score computed on all 2,000 concatenated test rows; ± is the sample standard deviation across seeds, which reflects the variability of the whole pipeline – fold assignment and training together – rather than model stability. Paired contrasts between these rows, under each metric and with their Holm-corrected p-values, are reported in Table 4. Under that variant M0 scores 0.567 against 0.572 on the official metric, while M1–M6 score identically under both: only the unconstrained arm predicts fields its own ancestors do not support.

table3_regimes

Method	Same-document	Document-disjoint	Δ	95% CI
M0	0.585	0.572	0.012	[0.004, 0.022]
Best calibrated projection	0.587	0.574	0.012	[0.001, 0.024]
Best valid-state decoder	0.590	0.576	0.015	[0.004, 0.027]

Same-document versus document-disjoint evaluation. The left column measures seen-report, unseen-paragraph generalisation and matches the competition distribution, since test and development draw on the same 49 reports; the right column measures generalisation to entirely unseen reports. Δ is the gap between two estimation targets and is not a bias estimate.

table4_contrasts

Metric	Contrast	Δ	95% CI	p_{Holm}
Weighted macro-F1 (official)	M1-M0	-0.001	[-0.006, 0.003]	1.000
	M3-M2	-0.001	[-0.006, 0.004]	1.000
	M4-M1	+0.000	[-0.005, 0.005]	1.000
	M6-M3	-0.007	[-0.017, 0.003]	0.643
	M6-M5	-0.009	[-0.017, -0.001]	0.135
Path-constrained wF1	M1-M0	+0.004	[0.001, 0.007]	**0.025**
	M3-M2	-0.001	[-0.006, 0.004]	1.000
	M4-M1	+0.000	[-0.005, 0.005]	1.000
	M6-M3	-0.007	[-0.017, 0.003]	0.482
	M6-M5	-0.009	[-0.017, -0.001]	0.108
Hierarchical F1 (hF)	M1-M0	+0.003	[0.001, 0.005]	**0.017**
	M3-M2	+0.002	[-0.001, 0.005]	0.302
	M4-M1	+0.004	[0.000, 0.007]	0.135
	M6-M3	+0.006	[0.000, 0.012]	0.151
	M6-M5	+0.004	[-0.001, 0.009]	0.302
Tuple accuracy	M1-M0	+0.035	[0.028, 0.043]	**0.001**
	M3-M2	+0.002	[-0.003, 0.007]	0.973
	M4-M1	-0.006	[-0.010, -0.002]	**0.028**
	M6-M3	-0.005	[-0.014, 0.004]	0.973
	M6-M5	-0.004	[-0.012, 0.004]	0.973

The five pre-specified contrasts of the analysis plan, each evaluated under every metric. Paired PDF-cluster bootstrap over 10,000 resamples on the document-disjoint protocol; the resampling unit is the source report, not the paragraph. Each metric is Holm-corrected as its own family of five rather than pooled: the contrasts were specified once, not once per metric. **Bold p_Holm marks a contrast that survives the correction; the bracketed intervals are uncorrected 95% percentile intervals and decide nothing on their own.** Of the five contrasts, none on Weighted macro-F1 (official), one on Path-constrained wF1, one on Hierarchical F1 (hF) and two on Tuple accuracy survive the correction at α=0.05. The official metric and tuple accuracy were named in the analysis plan in advance; the path-constrained variant and the hierarchical F1 were adopted after the primary analysis and are not pre-specified.

table5_metrics

ID	Weighted F1	C-wF1	hP	hR	hF	Tuple Acc.
M0	0.572	0.567	0.688	0.703	0.695	0.359
M1	0.571	0.571	0.707	0.690	0.698	0.394
M2	0.574	0.574	0.722	0.719	0.721	0.428
M3	0.574	0.574	0.723	0.722	0.722	0.431
M4	0.571	0.571	0.700	0.703	0.702	0.388
M5	0.576	0.576	0.721	0.730	0.725	0.430
M6	0.567	0.567	0.713	0.745	0.729	0.426

The same seven decision rules scored under each metric. C-wF1 is the official metric with ancestor-unsupported fields counted as false predictions; hP, hR and hF are the ancestor-based hierarchical precision, recall and F1, which are micro-averaged over path nodes rather than macro-averaged over classes. C-wF1 equals the official score for every rule whose output is legal by construction and falls below it only for the unconstrained one. The metrics disagree about the winner: M5 on the official metric and M6 on hF, which is the lowest-scoring rule on the official metric. A ranking is only meaningful stated together with the metric that produced it.

figure1_hierarchy